

Progress

AI-RAN GPU Isolation — Mechanism Deep-Dive

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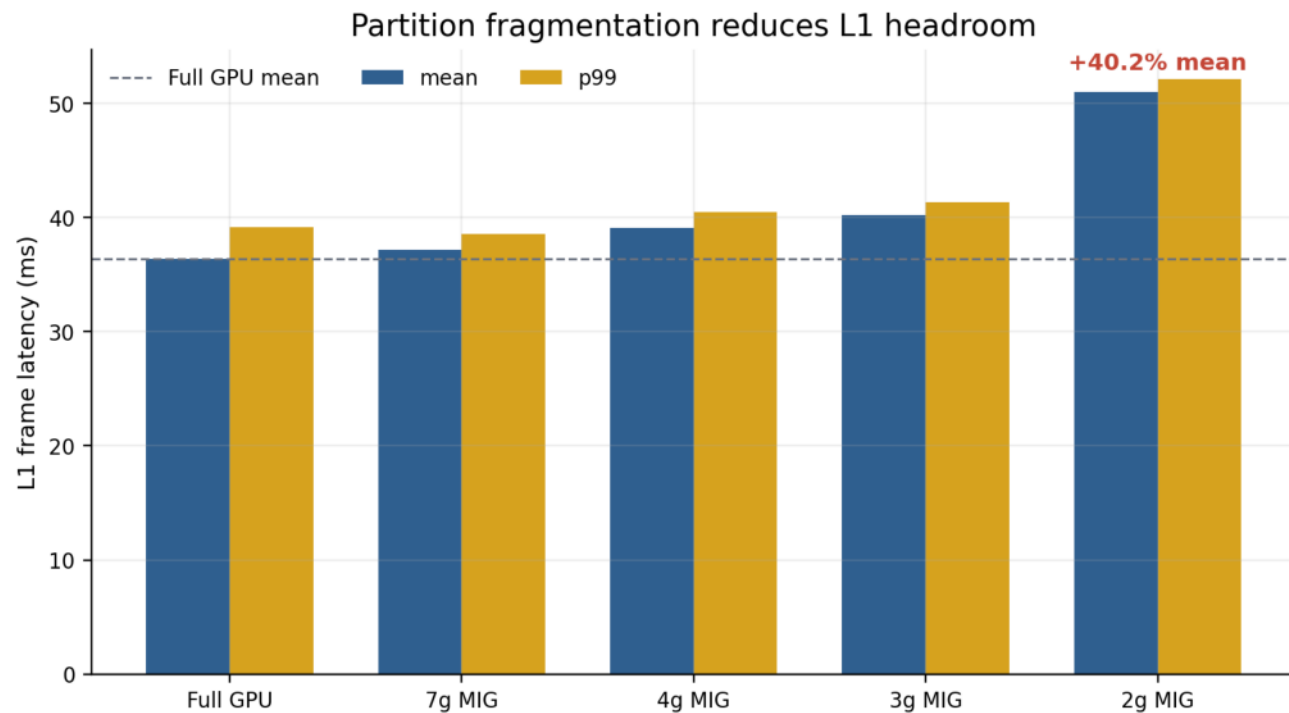
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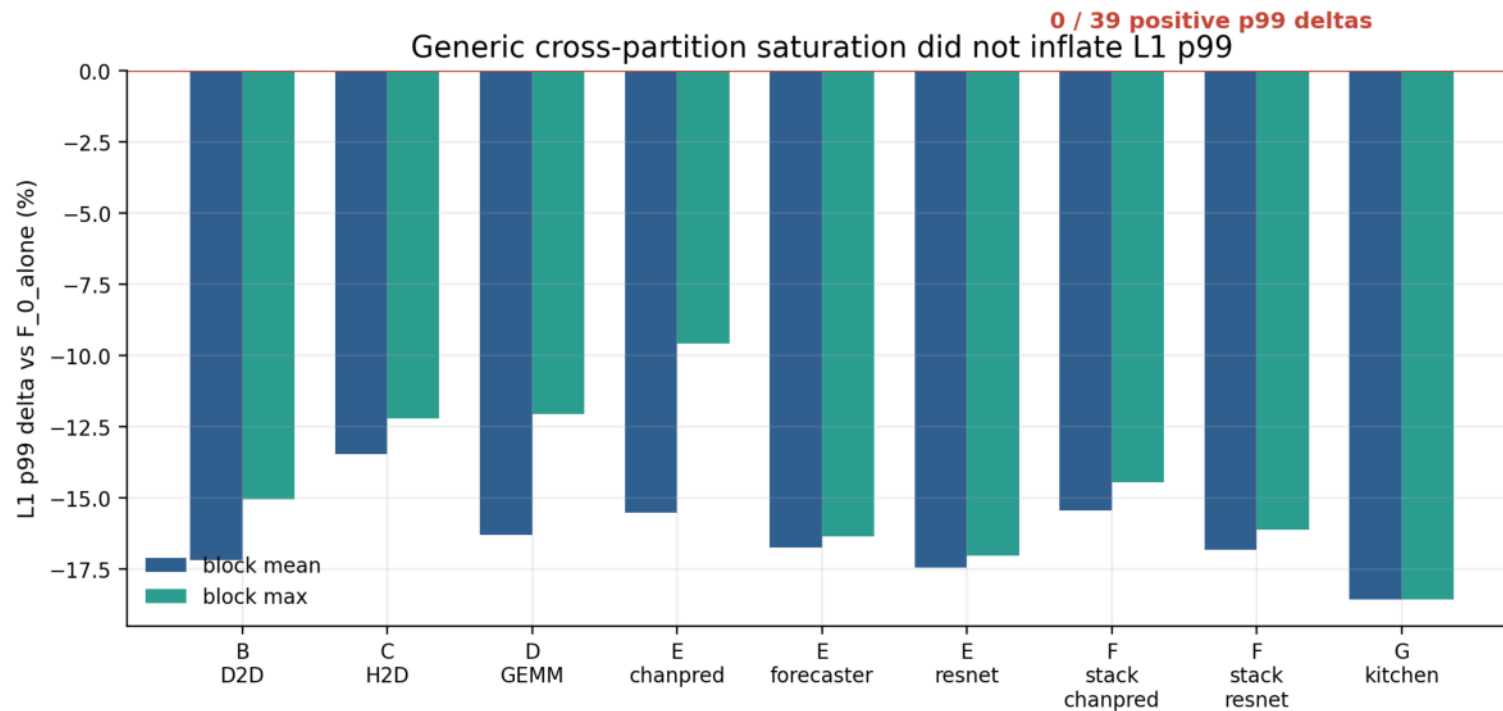
Today

- Continuation of Exp1-6 (5/29): MIG single-tenant overhead measured at +13-32 ms.
- Goal: Identify the hardware resource responsible for the L1 latency cost under AI co-tenancy.
- Setup: n=500-1000 per condition (25× the prior n=20), CloudLab d8545 (driver 550) + Perlmutter cross-validation.



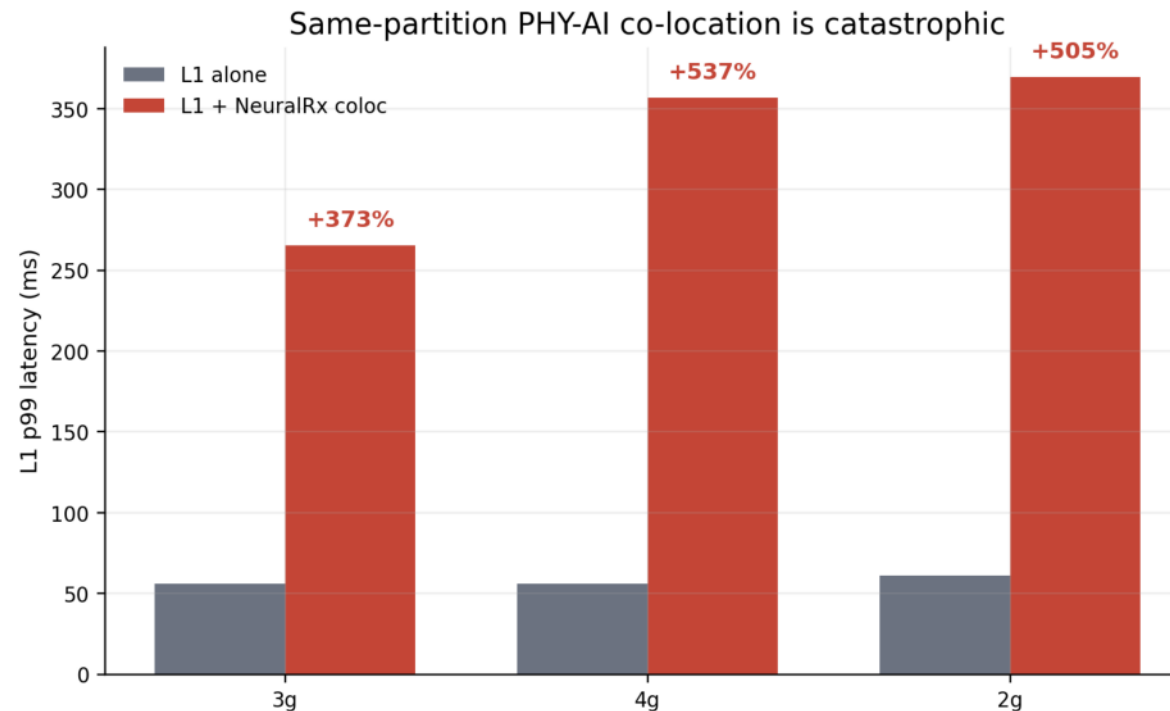
Generic Cross-Partition Saturation Does NOT Break MIG

- **Negative Result:** Cross-partition AI saturation (D2D, H2D, GEMM, ResNet, ChanPred, forecaster) keeps L1 within $\pm 10\%$ of alone baseline.
- **Stacking Does Not Help the Hypothesis:** ChanPred $\times 4$ and ResNet $\times 2$ stacking, plus a kitchen sink (cp+memcpy+gemm), also stay near alone baseline.
- **Result Across 39 Conditions:** 0 of 39 stressors produced a positive L1 p99 delta. Generic neighbor-partition saturation is not the cost driver.



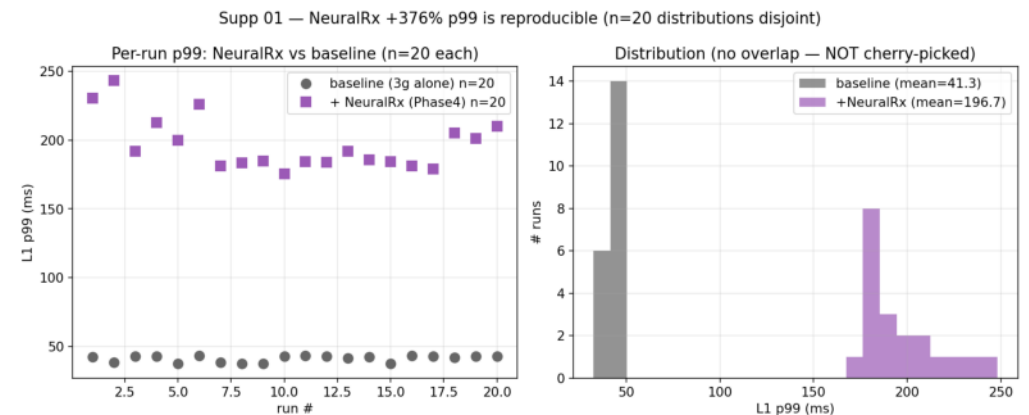
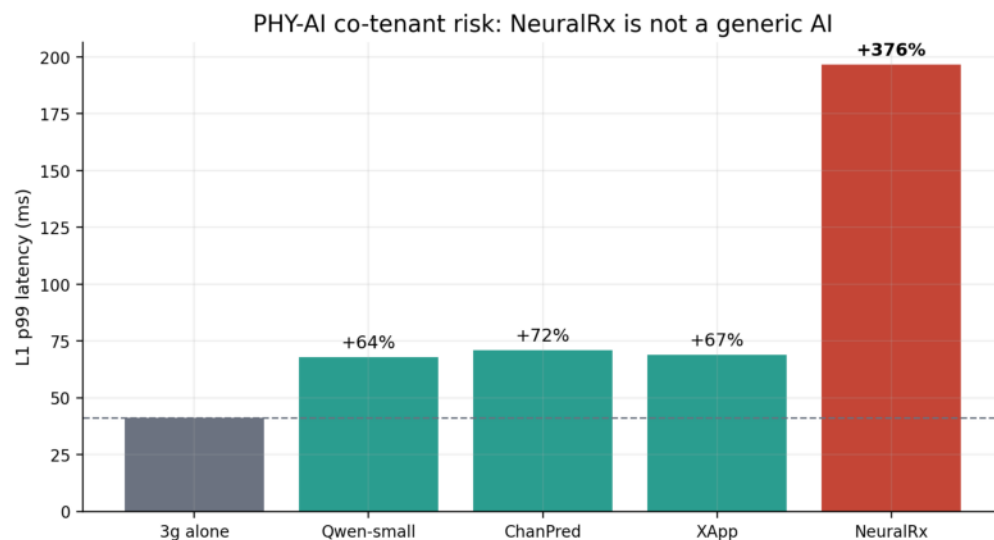
Same-Partition Co-location IS Catastrophic

- **8× Collapse:** Moving the same workload from cross-partition into the same partition as L1 inflates p99 from ~45 ms to ~357 ms.
- **Workload-Invariant:** chanpred, NeuralRx, qwen, xapp, and ResNet all converge to ~357 ms in coloc.
- **Cost Driver is Placement:** n=500/condition, std=9 ms. The cost is set by where AI is placed, not by what AI is doing.



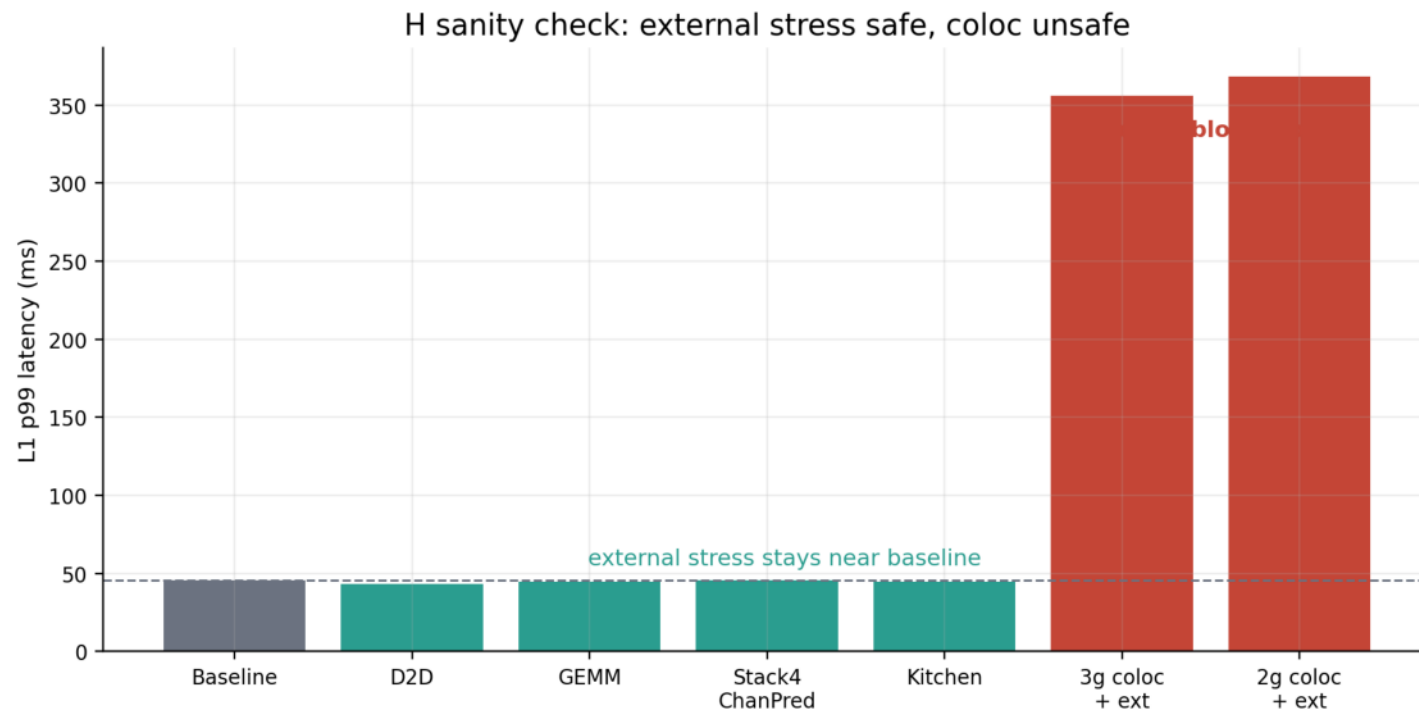
NeuralRx is Uniquely Dangerous Even Cross-Partition

- **PHY-AI Breaks the Negative Result:** NeuralRx in a separate partition still inflates L1 p99 by +376% (41 → 197 ms); ChanPred/xApp/Qwen only +60-72%.
- **Not Cherry-Picked:** Across n=20 runs, baseline (37-44 ms) and +NeuralRx (175-243 ms) form completely disjoint distributions.
- **Cause:** NeuralRx shares cuPHY-style copy/convert pattern with L1, producing contention even across MIG partition boundaries.



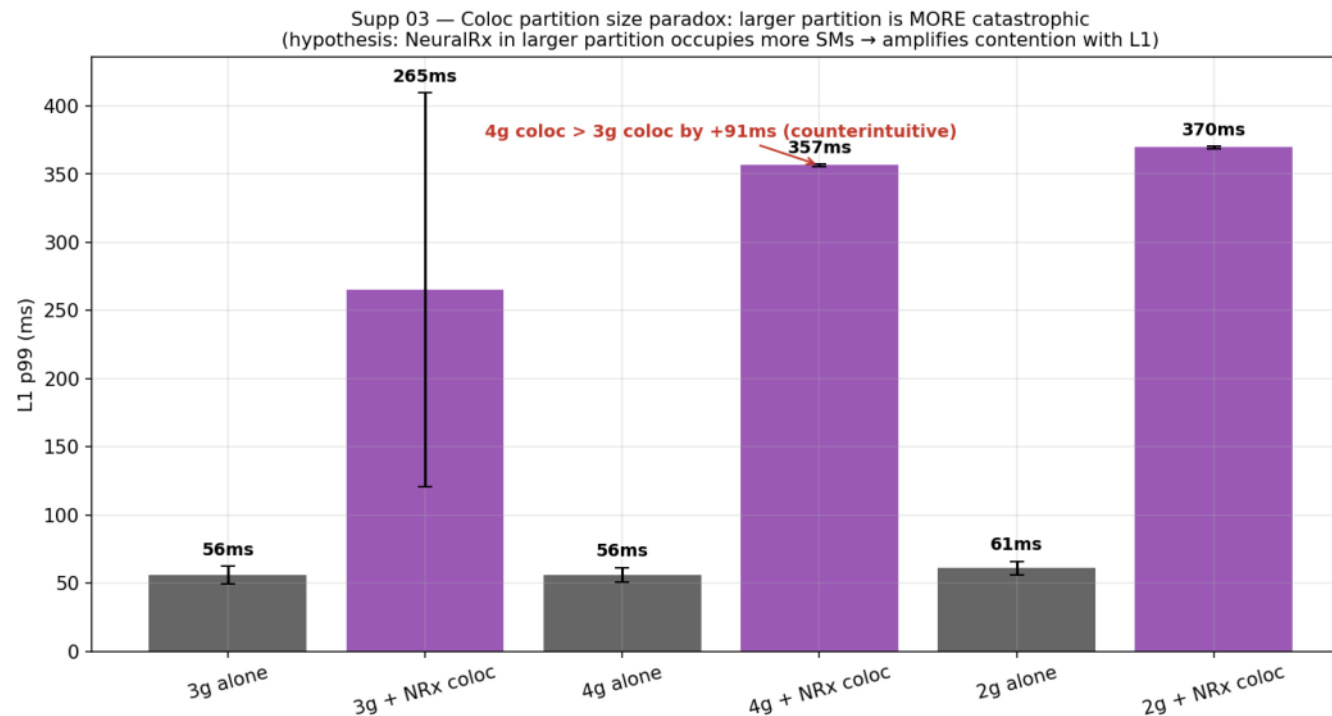
Placement is a Binary Decision

- **No Soft Transition:** Identical chanpred workload reads 44 ms cross-partition vs 357 ms same-partition.
- **One-Switch Penalty:** A single placement misconfiguration produces the full collapse; no intermediate values exist.
- **Operational Consequence:** Latency cost is binary, not gradual. Admission control must enforce placement explicitly.



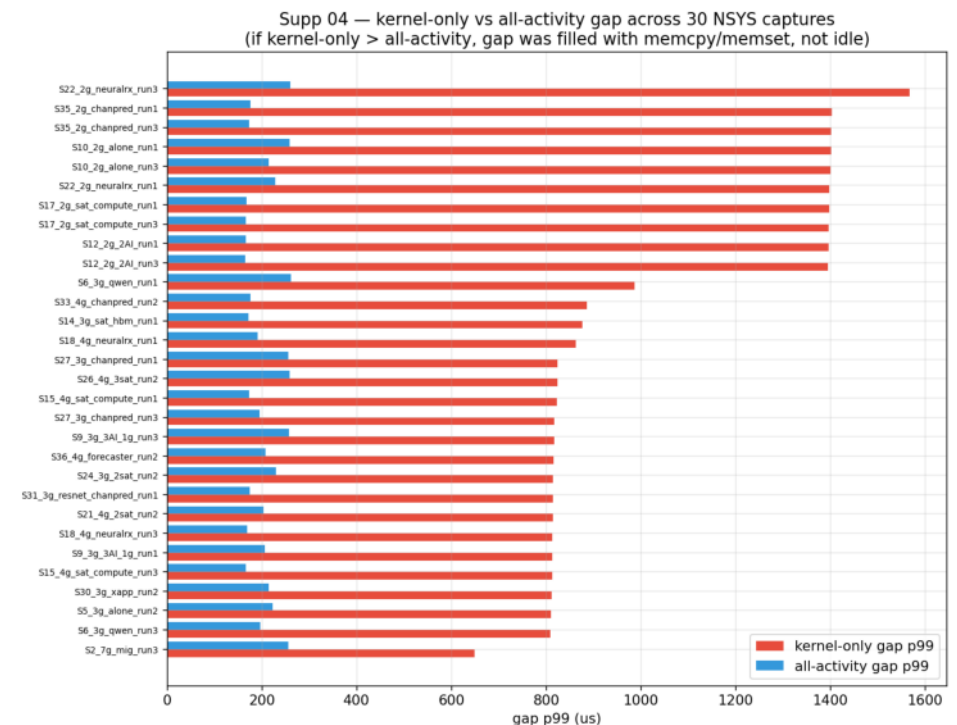
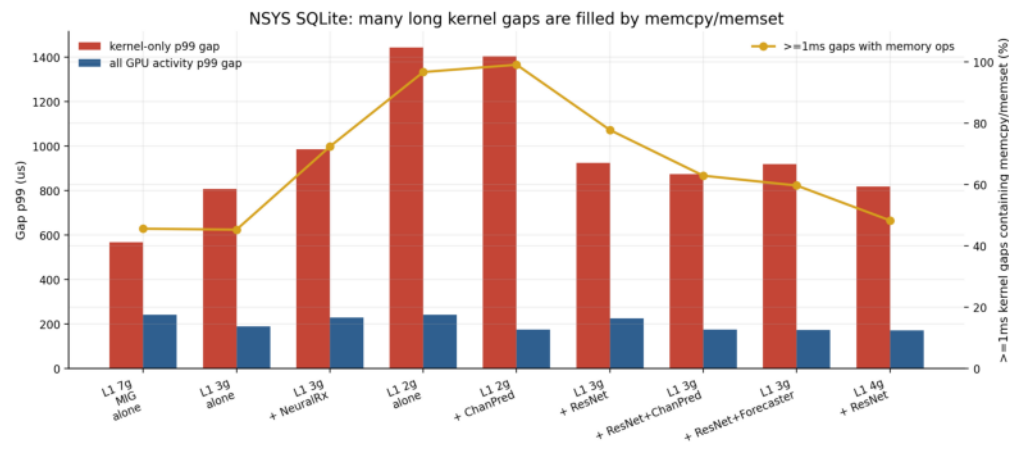
Larger Partition = MORE Catastrophic

- **Counter-Intuitive Outcome:** 4g co-location (371 ms) is worse than 2g co-location (369 ms).
- **Resource Allocation Does Not Mitigate:** Adding SMs and HBM through partition sizing does not reduce the collapse.
- **Bottleneck is Chip-Global:** The contended resource lies outside the per-partition abstraction MIG provides.



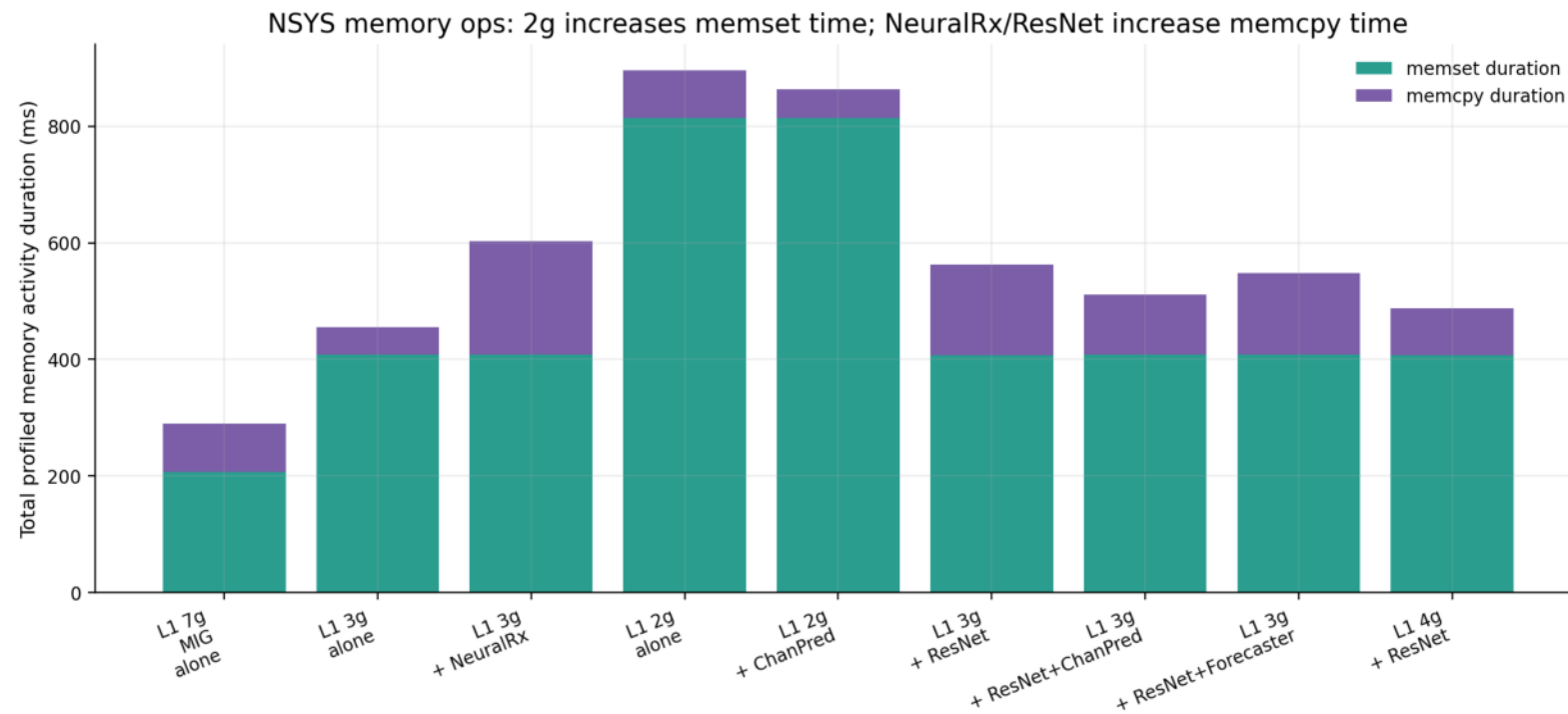
Mechanism Layer 1 — Inter-Kernel Gaps Cluster at Memory-Op Boundaries

- **Kernel Duration Unchanged:** Per-kernel execution time is identical under alone and co-location; only inter-kernel gaps grow.
- **Gaps Are Not Idle:** Inter-kernel gaps coincide with memcpy and memset boundaries.
- **Robust Across 30 NSYS Captures:** kernel-only gap p99 > all-activity gap p99 in every condition — the gap is filled with memory ops, not idle time.



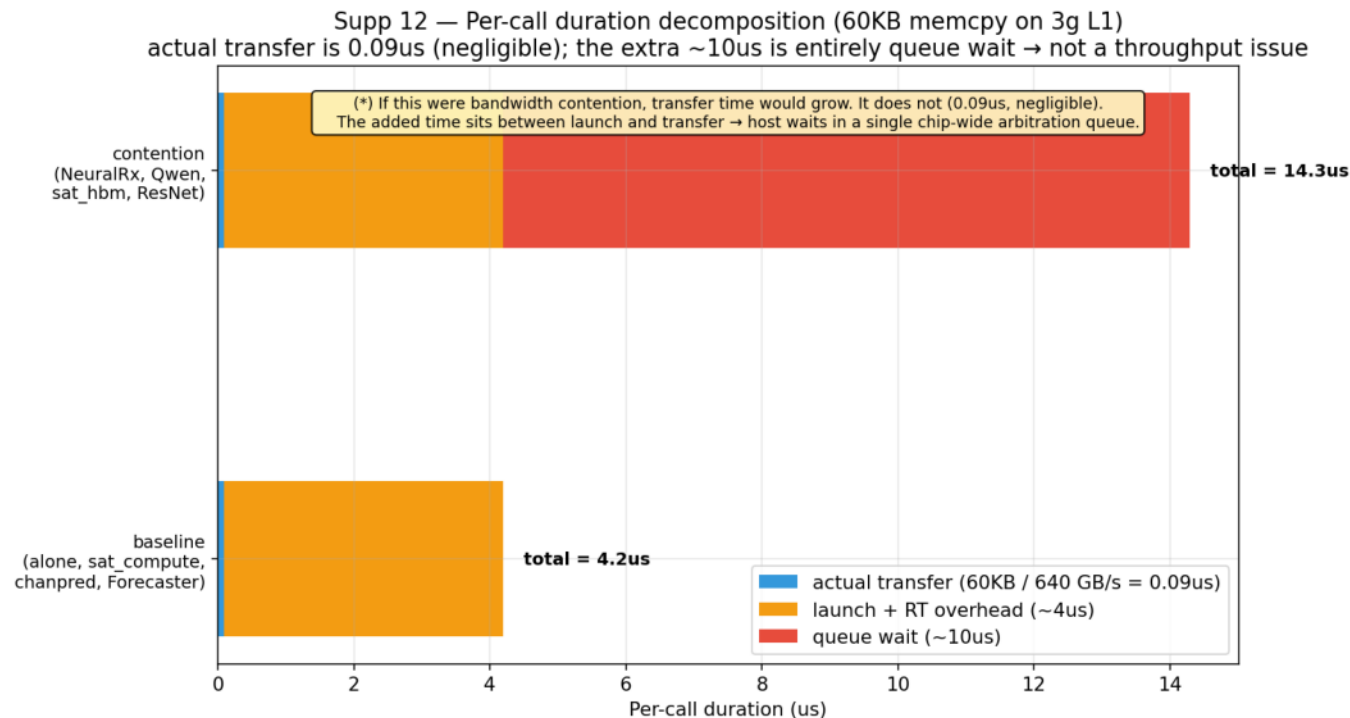
Mechanism Layer 2 — Op Count Stable, Duration Inflates

- **Call Count Invariant:** memcpy and memset issue identical call counts (5,778 each) under alone and co-location.
- **Per-Call Duration Grows 133%:** memcpy average duration rises from 4.2 μ s to 9.8 μ s for the same number of operations.
- **Excludes Demand-Driven Hypotheses:** L1 is not asked to do more work; each operation simply takes longer.



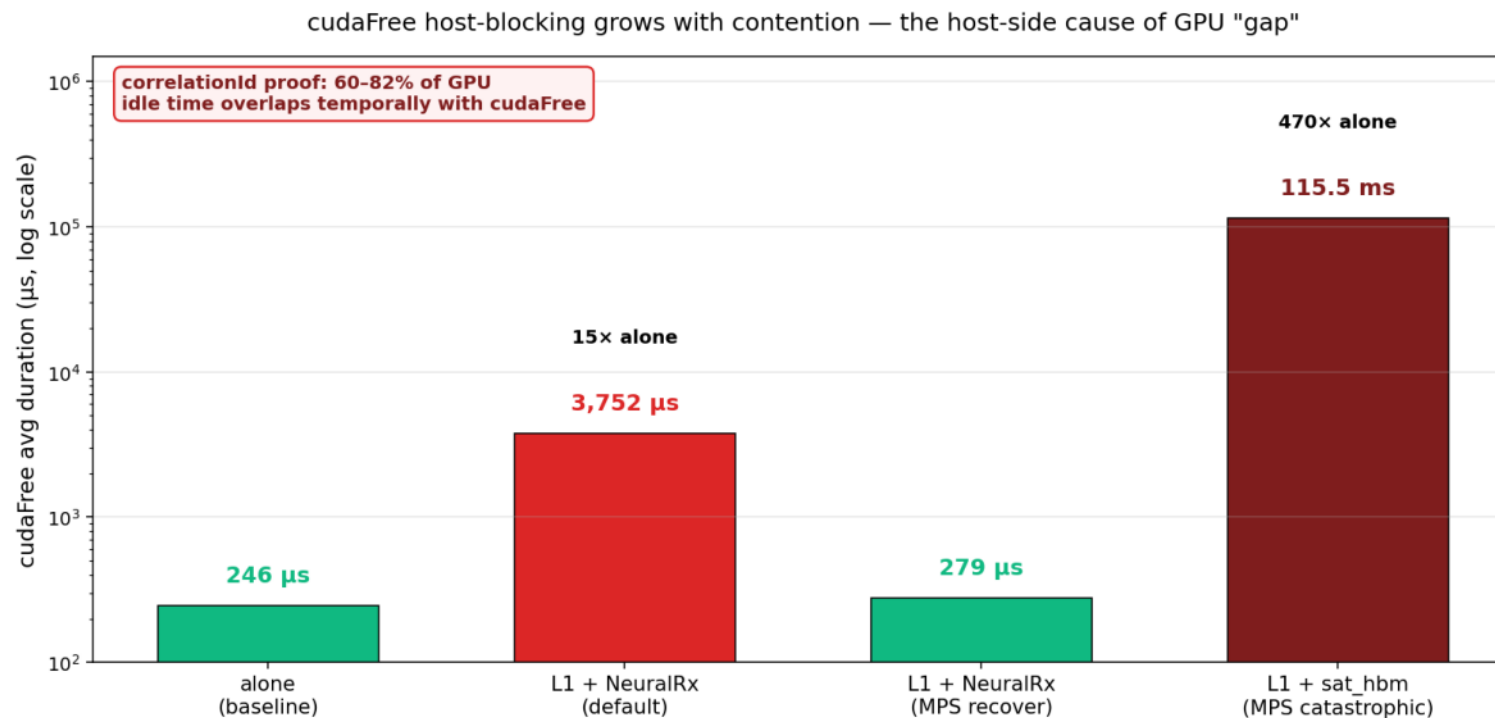
Mechanism Layer 3 — Time Decomposition Proves Queue Wait, Not Throughput

- **Actual Transfer is Negligible:** $60 \text{ KB} / 640 \text{ GB/s} = 0.09 \mu\text{s}$. Bandwidth saturation cannot account for the inflation.
- **Baseline Decomposition:** $4.2 \mu\text{s}$ alone = launch + runtime overhead; transfer contributes essentially nothing.
- **Contended Decomposition:** $14.3 \mu\text{s} = 4.2 \mu\text{s}$ baseline + $\sim 10 \mu\text{s}$ queue wait. The slow mode is queue wait by construction.



Mechanism Layer 4 — cudaFree Host-Blocking is the Direct Cause

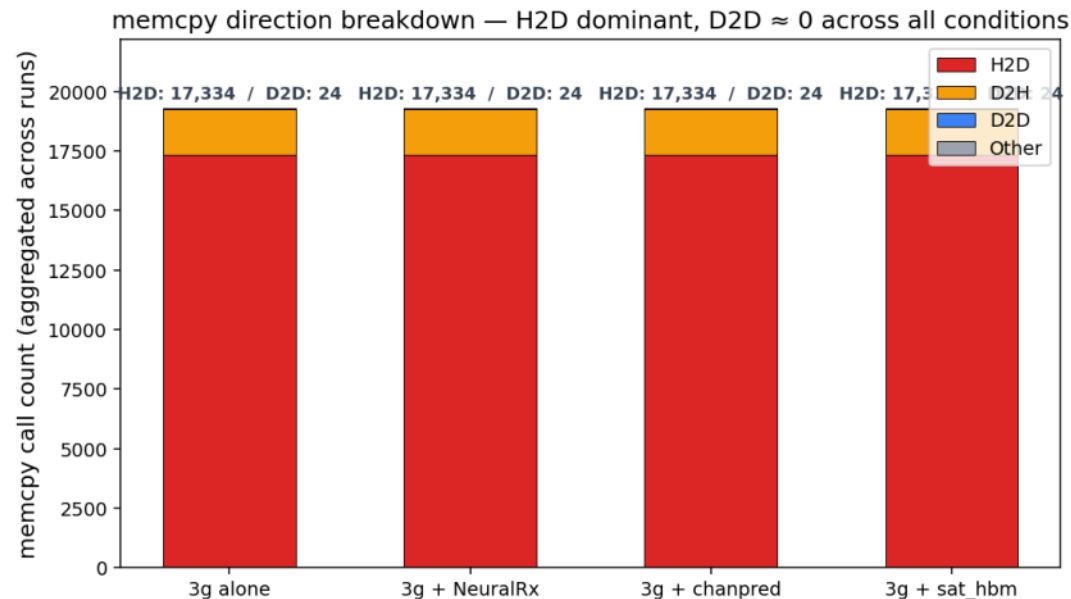
- **Host-Side cudaFree Inflates with Contention:** cudaFree average grows from 246 μs (alone) to 3,752 μs (NeuralRx default, 15 $\times$) to 115,506 μs (sat_hbm MPS, 470 $\times$).
- **Same Call Recovers Under MPS:** cudaFree returns to 279 μs under MPS + NeuralRx — host blocking resolves when the device is freed.
- **Causal Proof via correlationId:** 60-82% of GPU idle gap time aligns temporally with cudaFree blocking on the host.



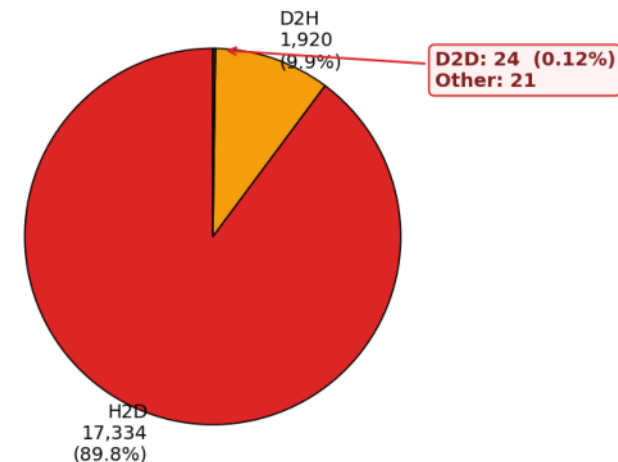
Mechanism Layer 5 — Queue Located at Chip-Wide PCIe/DMA Copy Engine

- **Direction Breakdown:** 89.8% H2D, 9.9% D2H, 0.12% D2D across all measured memcpy calls.
- **Not the HBM Controller:** D2D ≈ 0 rules out the HBM controller, which would carry D2D traffic.
- **Single Shared Resource:** The PCIe/DMA copy engine is one physical unit serving the entire chip; partitioning cannot isolate it.

Queue location proof — memcpy direction: 100% H2D, D2D ≈ 0
→ Queue is at chip-wide PCIe/DMA copy engine (not partition-isolated HBM controller)

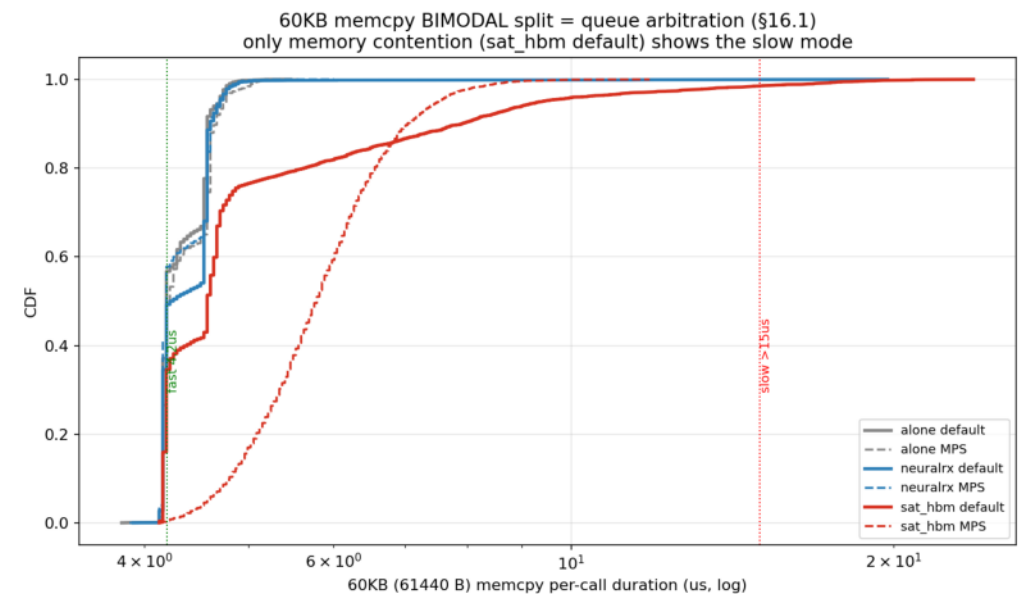
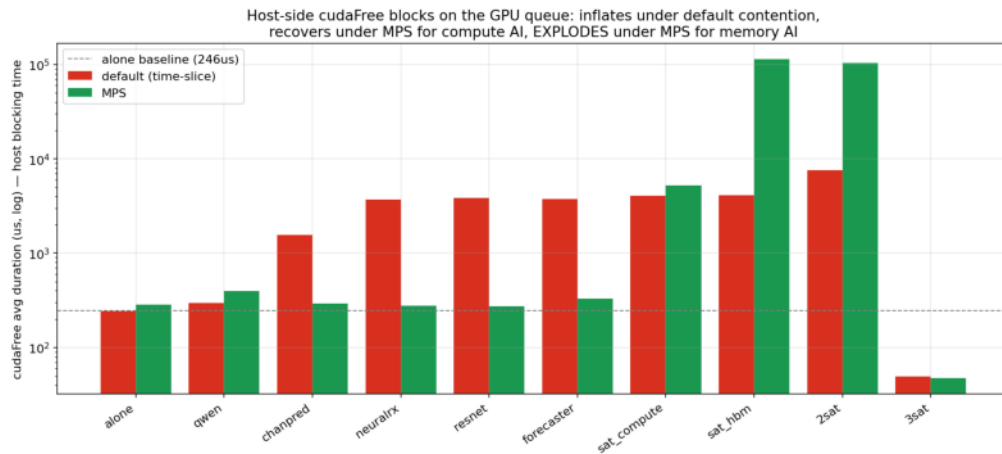


3g alone — proportional breakdown



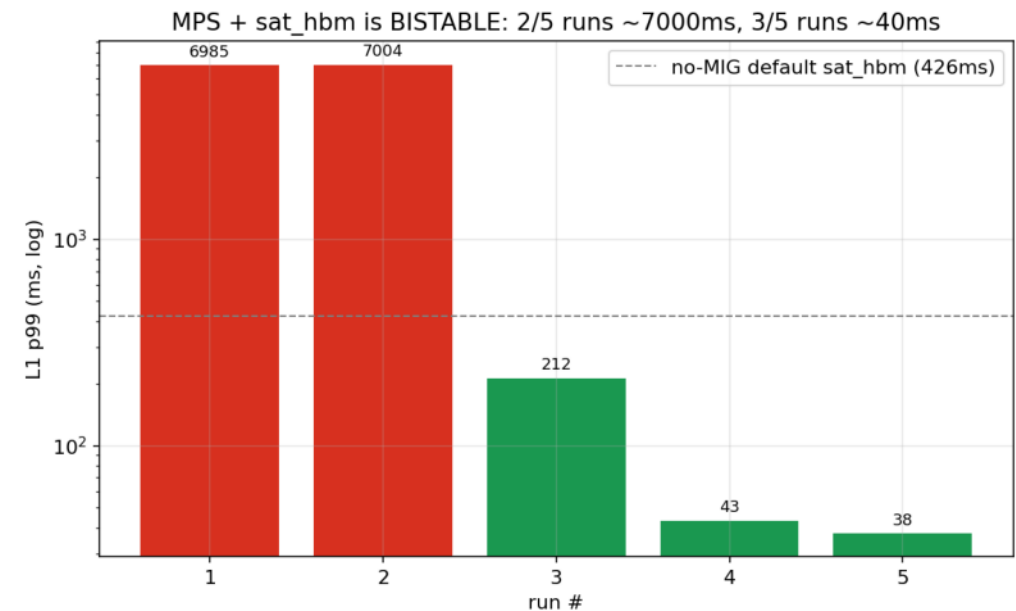
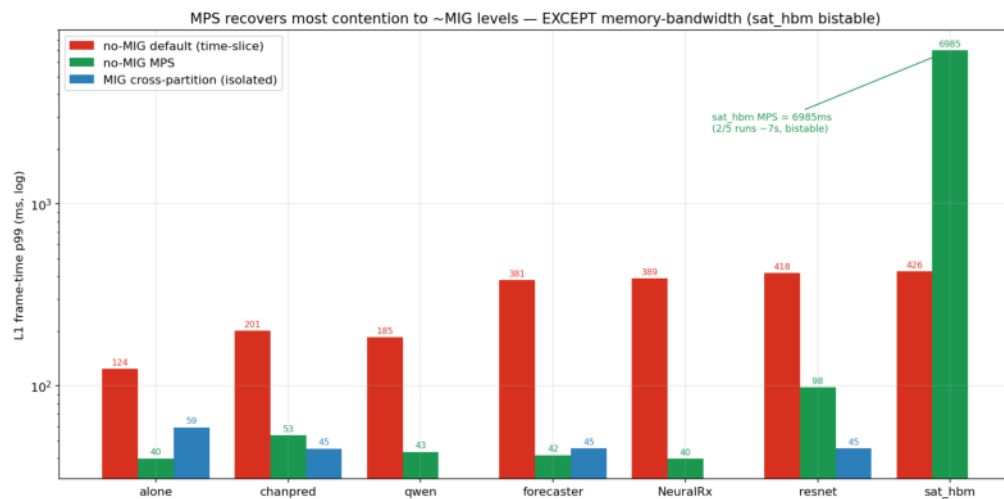
Cross-Platform Mechanism Validation

- **Bimodal Signature Reproduced:** Perlmutter no-MIG shows the same 60 KB memcpy bimodal split ($4.2 \leftrightarrow 16.8 \mu\text{s}$).
- **cudaFree Scaling Matches:** cudaFree average grows 15 $\times$ under NeuralRx default and 469 $\times$ under MPS + sat_hbm.
- **Conclusion:** The mechanism is NVIDIA-stack-wide. It is not specific to MIG or to the CloudLab driver.



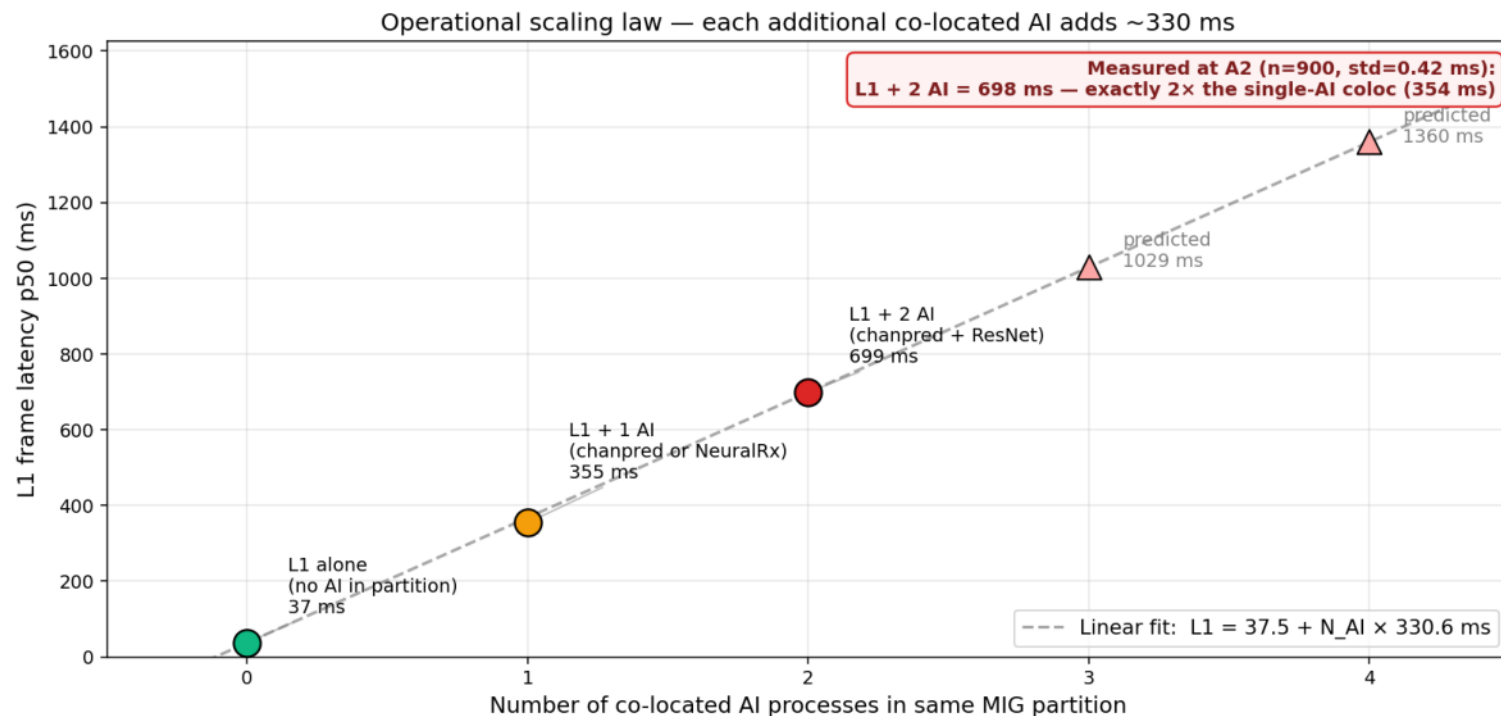
MPS Recovers Compute AI, Catastrophic for Memory AI

- **Compute AI Recovers:** NeuralRx 389 → 40 ms (below MIG cross-partition 197 ms); forecaster 381 → 42 ms; Qwen 185 → 43 ms.
- **Memory AI Becomes Worse:** sat_hbm 426 → 6,985 ms under MPS (bistable across runs).
- **Same Mechanism, Different Trigger:** MPS routes through the same cudaFree path; memory-bound workloads starve the device and re-engage the failure mode.



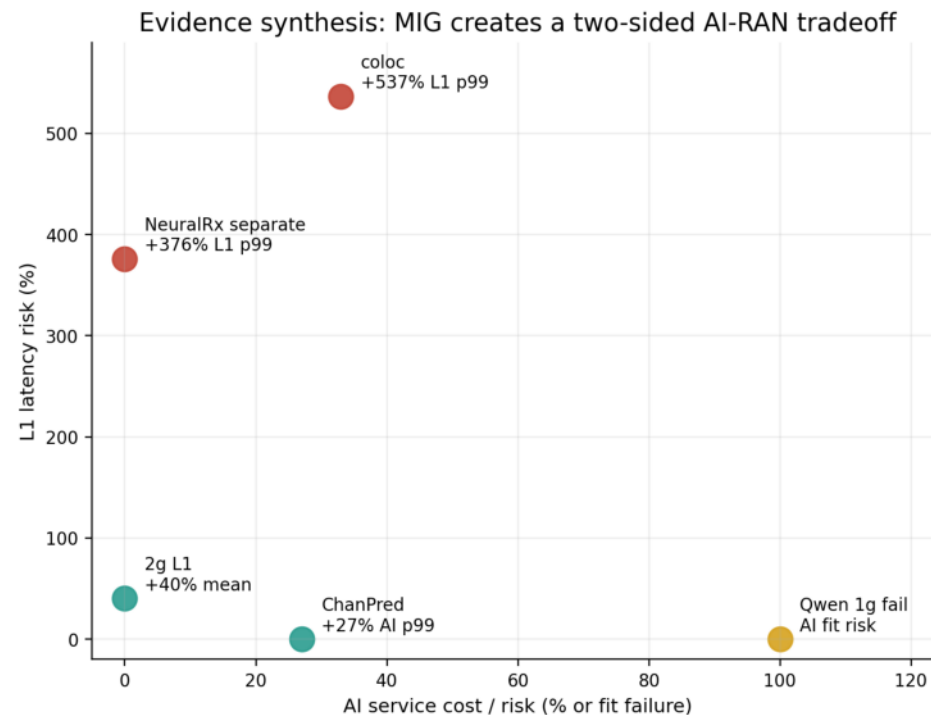
Operational Scaling Law — Each AI Adds ~330 ms

- **Linear Cost Model:** $L1 = 37.5 + N_AI \times 330.6$ ms when AI processes share the same MIG partition.
- **A2 Validates Slope:** L1 + 2 AI measures 698 ms (n=900, std=0.42 ms) — exactly 2× the single-AI co-location (354 ms).
- **Multi-Tenant Mistakes Compound:** Predicted 1,029 ms at 3 AI and 1,360 ms at 4 AI. Operational error scales linearly, not as a single step.



Cost Decomposition — Two Hardware-Level Costs

- **Structural Cost (memset):** Tied to partition HBM bandwidth share (2/7, 3/7, 4/7); selectable through partition sizing.
- **Contention Cost (memcpy):** Tied to the chip-wide PCIe/DMA copy engine; not isolable by any partition choice.
- **Unified Across Three Platforms:** MIG, no-MIG, and MPS each avoid one cost while remaining exposed to the other.



Design Rules & Next Steps

- **Placement Enforcement is Mandatory:** L1 and AI must reside in different MIG partitions; one misplacement produces the full 8× collapse.
- **Partition Sizing Serves Only L1 Budget:** Sizing has no mitigating effect under co-location; choose for L1 SM/HBM headroom only.
- **AI Workload Classification is Required:** Memory-bound AI cannot be safely co-located even under MPS.
- **MIG $\neq$ Chip-Wide Isolation:** MIG isolates per-partition SM/HBM but not the chip-global PCIe/DMA copy engine.

Thank You
